

UNIT 4

DATA AND ANALYSIS

9TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

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DATA

- Data is an information or facts that we gather about something. It can be represented by numbers, measurements, descriptions, sounds or pictures.
- Here are two examples of data. In a science experiment when you record the temperatures at different times, that temperature values are data. If you conduct a survey of your classmates and get to know how many of them like Mathematics, it will be called data.

DATA ANALYTICS

- Data Analytics refers to the process of carefully studying data to identify patterns and draw conclusions.
- To analyze data, we use mathematical calculations, statistical techniques, charts, or other tools to understand data.
- For example, after recording hourly temperature data, you can create a graph to see how it changed over time. From the graph, you conclude that it got warmer as the day went on — that is the result of your data analysis.
- So, data is the information we collect, and data analytics is how we observe it to get meaningful information. Data analytics can be quantitative (numeric – height, temperature, marks) or qualitative (descriptive - color, gender, taste)

DATA SCIENCE

- Data Science refers to an interdisciplinary field of mathematics, statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it.
- The insight result drawn can be used for making informed decisions to solve real world problems e.g., medical, education, scientific research, and business etc.

CONCEPTS OF DATA SCIENCE

Data

A data is an information collected from different sources. It could be structured in the form of tables or unstructured in the form of audio, video, tweets, pdf files etc.

Dataset

A collection of related data organized in a structured way which is ready for the project of data science.

Statistics and Probability

Statistics is the analysis of the frequency of past events and probability is to predict the likelihood of future events. Data scientists use statistics and probability to find patterns and trends in the data.

CONCEPTS OF DATA SCIENCE

Mathematics	Mathematics is a fundamental part of data science which helps to solve problems, optimize the model performances, and interpret huge complex data into simple and clear results, for decision making.
Machine Learning	Machine Learning is a branch of Artificial Intelligence (AI) that allows computers to learn from data without being explicitly programmed.
Deep Learning	Deep learning is the subset of Machine learning, with emphasis on the simulation or imitation of human brain's behavior by using artificial neural networks.

CONCEPTS OF DATA SCIENCE

Data Mining	Data mining is the process of discovering patterns and relationships in existing datasets. The usage of techniques and tools is limited in data mining as compared to data science.
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Data Visualization	Data visualization is the graphical representation of data using common charts, plots, infographics, and animations.
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Big Data	Big data refers to handling large volumes of data. The huge size of data provides more opportunities for machine learning and provides better results.
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CONCEPTS OF DATA SCIENCE

Predictive Analysis

Predictive analysis is the use of data to predict future trends and events based on historical data.

Natural Language Processing (NLP)

It is the study of interaction between human language and computers. The common uses of NLP are chatbots, language translators and sentiment analysis.

SCOPE AND APPLICATION OF DATA SCIENCE

- Data science is used for a wide range of applications which includes predictive analytics, data visualization, recommendation systems, sentiment analysis, fraud detection, and decision-making in various industries like healthcare, finance, marketing, and technology.
- Apart of them, various business problems can be solved through data science, some of them are as follows:
 - *To decide the best routes for shipping of goods or passenger airplanes*
 - *To choose the best product among many, which one to buy A or B*
 - *To foresee delays for flight/ship/train etc. (through predictive analysis)*
 - *To create promotional offers (which products are more popular than others)*
 - *To find the best suitable time to deliver goods to reduce cost*
 - *To forecast next year's revenue for a company*
 - *To analyze health benefit of physical training programs*
 - *To predict some fore coming event like who will win elections*

BUSINESS PROBLEMS AND DATA SCIENCE

Industry: Data science helps make **data-driven decisions** by analyzing historical data, predicting future trends, improving marketing, and enhancing quality control.

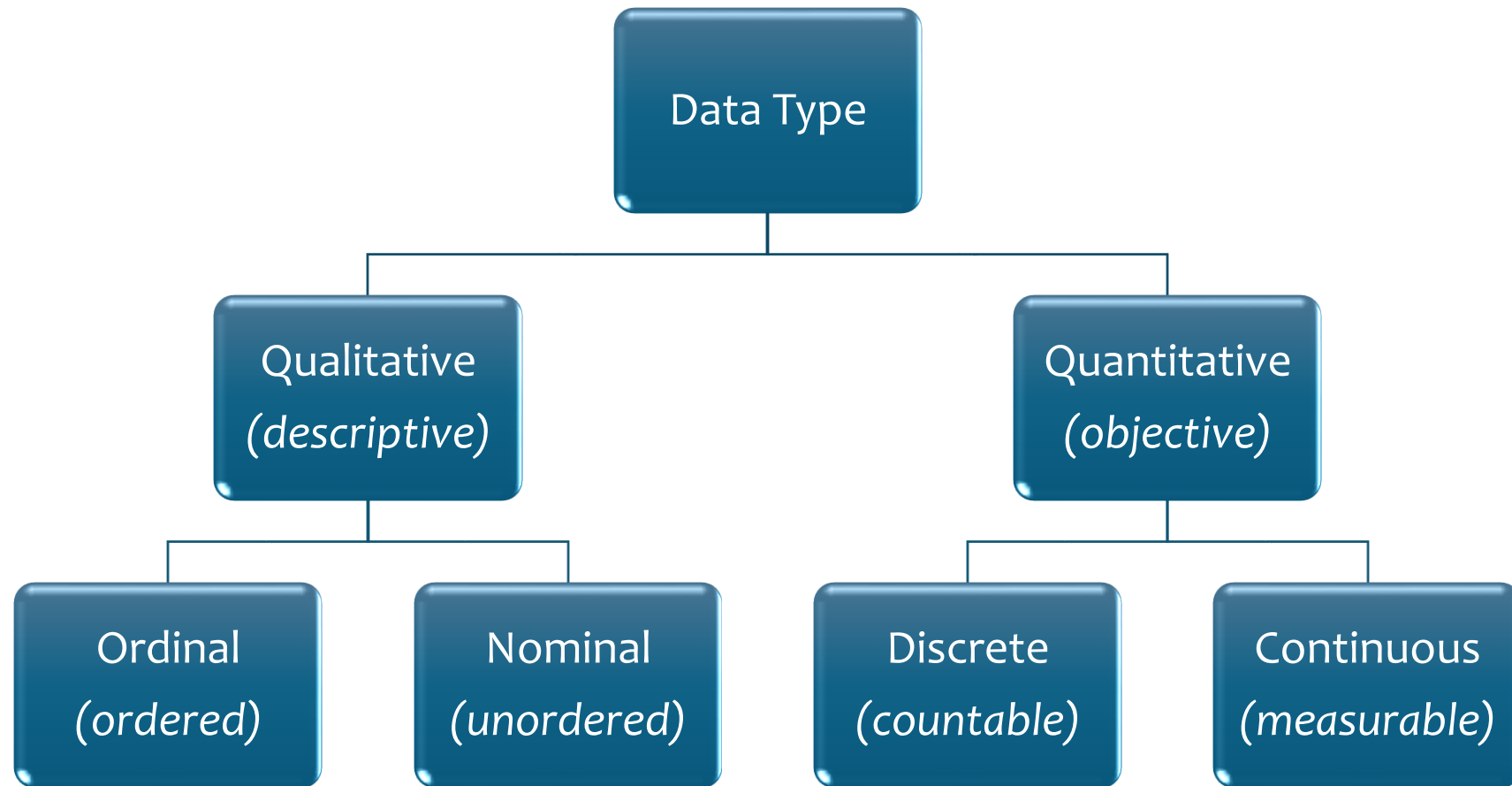
Consumer Goods: It helps optimize inventory according to demand forecasting for particular social groups, communities, and demographics.

Logistics Companies: These companies use it for route optimization, demand forecasting, real-time tracking, load balancing, carrier selection, cost reduction, and global trade optimization.

Stock Markets: Data science helps with algorithmic trading, market sentiment analysis, volatility predictions, quantitative analysis, machine-learning based trading, market surveillance, and risk management.

E-commerce: It supports recommendation systems, customer segmentation, shopping cart analysis, fraud detection, supply chain optimization, and customer sentiment analysis.

DATA TYPES IN DATA SCIENCE



DATA TYPES IN DATA SCIENCE

- **Qualitative or Categorical data** describes a group of objects that can be labeled according to some category. For example, data including colors, places, etc. It is further subdivided into two types:
 - **Ordinal Data:**
Ordinal data sees a specific ranking or order. Such as in (1st, 2nd, 3rd) economic status (Low, Mid, High), or military rank (Major, Major General etc.)
 - **Nominal Data:**
Nominal data does not have any order and can be labelled into mutually exclusive categories. For example, if we consider the categories of transportation as vehicle (car, bus, van) , gender (male, female), city (Karachi, Islamabad), color (brown, green), employment status (employed, unemployed) etc.
- **Quantitative or Numerical data** deals with numeric values. Examples of numeric data are height, weight, number of students in a school, fruits in a basket etc. Quantitative data can be further divided into two types:
 - **Discrete Data:** *It includes countable data form which can only take certain values and cannot be further subdivided into smaller units. For example, the number of product reviews (5), ticket sold (250), computers in lab (20).*
 - **Continuous Data:** *It refers to the unspecified uncountable number of possible measurements in points. For example, daily wind speed (12mph), weight of newborn babies (1.5kg), freezer's temperature (23.43 degree) etc.*

SOURCES OF DATA

- **Data Collection** is the first step in analyzing data for predictive analysis and decision making, done through reliable sources.
 - *Primary Data* is collected directly by **questionnaires, surveys, interviews, experiments, and recording observations**.
 - *Secondary Data* is collected from data that was **previously recorded** from primary sources.

Sources of Data:

- **Website:** Collecting tweets regarding a topic or thread.
- **Surveys:** Collecting firsthand data about an event, movie, or anything else.
- **Sensors:** Collecting seismic data about changes under the earth that cause earthquakes.

DATASET AND DATABASE

- A **dataset** is a structured or organized collection of data in form of table which is ready for data science project. Example (Excel Workbook, SQL Table)
- A **database** is a storage where data is stored and can be accessed electronically for further operations – CURD (Create, Retrieve, Update, Delete). Example (DB of Telcom Company, School Record).
- To perform CURD in a database, we need a **Database Management System (DBMS)**. There are different DBMS types depending on the database:
 - **Relational databases** store data in tables — managed by MySQL, Oracle, MS-Access, IBM Db2. These are most used in Data Science for tabular data.
 - **Non-relational databases** store data as key-value pairs, column families, or graphs — managed by MongoDB and Cassandra. These are also called NoSQL DBMS.

DATA COLLECTION IN DATA SCIENCE

Data Collection is the process of collecting information from relevant sources to find a solution to the given statistical inquiry. Collection of Data is the first and foremost step in a statistical investigation. Two Data collection methods are:

Primary data collection methods

It involves the collection of original data directly from the data source or via direct interaction with the respondent. A respondent is a person from whom the statistical information required for the inquiry is collected. Some common primary data collection methods are as follows:

- i. Surveys and Questionnaires
- ii. Interviews
- iii. Observations
- iv. Experiments
- v. Focus groups
- vi. Sensors
- vii. IoT devices
- viii. Biometric devices

Secondary data collection methods

It involves data collection using existing data collected by someone else for some purpose. Such data is usually available in the form of published material like research papers, books, websites etc. Some common secondary data collection methods are as follows:

- i. Published sources
- ii. Online databases
- iii. Government and institutional records
- iv. Surveys and Questionnaires conducted in the past
- v. Social media data/posts
- vi. Publicly available data
- vii. Past research studies

DATA STORAGE

- After data collection, **effective storage** is essential for managing and analyzing large volumes of data. There are various storage methods depending on the **nature of the data**.

Common data storage methods:

- **Relational Databases** – Store data in **tables** (rows and columns) with fixed relationships. Use **SQL**. Example: MySQL, Oracle
- **NoSQL Databases** – Store data in **flexible formats** like key-value pairs, documents, or graphs. Good for **unstructured data**. Example: MongoDB, Cassandra.
- **Data Warehouse** – A **central storage system** that collects data from multiple sources for **analysis and reporting**. Example: Amazon Redshift.
- **Distributed File Systems** – Store data across **multiple machines** so it can be accessed and processed in parallel. Example: HDFS (Hadoop).
- **Cloud-Based Data Storage** – Stores data on **remote servers** accessed via the internet. Example: Google Drive, AWS S3.
- **Blockchain** – A **decentralized, secure ledger** where data is stored in **linked blocks** that cannot be altered. Example: Bitcoin.

DATA TERMS

Data Visualization

Data visualization is graphical representation of data to get meaningful insight, trends, and patterns from data. The visual elements which help in data visualization are charts, graphs, maps, figures, and dashboard etc.

Summary Statistics

Summary Statistics is information about the data in a sample that helps understand the values better. It may include the **total number of values, minimum, maximum, mean, and standard deviation** of a data collection. It helps identify **trends, outliers, and distribution** of values in a dataset.

BIG DATA

Big Data contains greater **variety**, arriving in increasing **volumes** and with more **velocity** — known as the **three Vs**. It refers to **larger, more complex datasets** from new sources that traditional software **cannot manage**. These massive volumes can solve business problems that were previously difficult to handle.

The Three Vs of Big Data:

- **Volume** – The **amount of data**. Deals with huge volumes of low-density, unstructured data — ranging from **terabytes to petabytes**.
- **Velocity** – The **speed of data**. Data arrives at a **fast rate**, often streaming directly into memory for **real-time evaluation and action**.
- **Variety** – The **different formats and types** of data. Includes **unstructured** (text, images, videos) and **semi-structured** (JSON, XML) data that require preprocessing for insights.

THE HISTORY OF BIG DATA

The term big data emerged in the early 2000s as a term to describe exponential growth of data. Around 2005, people began to realize just how much data users generated through Facebook, YouTube, and other online services. In 2005 a tool called Hadoop (an open-source framework created specifically to store and analyze big datasets) was developed, which helped store and manage huge data.

With the advent of the Internet of Things (IoT), more and more objects and devices are connected to the internet, gathering data on customer usage patterns and product performance. The emergence of machine learning has produced still more data. The analysis of this huge data provides business insight for optimized decision making.

ADVANTAGES AND BENEFITS OF BIG DATA

Big Data helps individuals, organizations, and businesses optimize and create cost-effective solutions. Some key advantages:

- **Product Development** – Companies use big data to understand customer needs and wants, anticipate demand, and build predictive models for new products.
- **Predictive Maintenance** – Uses data analysis to predict when equipment or machinery may fail, catching potential issues before they happen.
- **Customer Experience/Satisfaction** – Gathers data from social media, web visits, and call logs to improve customer satisfaction.
- **Fraud and Compliance** – Detects unusual patterns and anomalies, making it an effective tool to catch fraud and enhance cybersecurity.

BIG DATA CHALLENGES

Since there are many advantages of big data, businesses encounter many challenges of big data. Some of them are as follows:

- **Data Quality:** *Poor quality of data may lead to errors, inefficiency, and misleading insight after data analysis.*
- **Data Security and privacy:** *It is difficult to manage the protection and privacy of massive datasets to prevent unauthorized access.*
- **Rapid growth of data:** *Making systems that can handle more and more data as it keeps on growing without slowing down is challenging.*
- **Big data tool selection:** *Ensuring compatibility and seamless interaction between different big data tools and platforms.*
- **Data Integration:** *To create harmony among diverse data formats and structures is a difficult task.*

APPLICATION OF BIG DATA IN BUSINESS

Big data helps companies make **better business decisions** by analyzing large volumes of data and discovering **hidden patterns**. Key domains include:

Healthcare – Wearable devices and sensors collect **patient data** in real-time into electronic health records. Big data helps **predict epidemic outbreaks, provide real-time alerts, prevent serious conditions, determine best treatments, track drug side effects, and forecast health risks.**

Media and Entertainment – Media houses use big data to **target audiences** by predicting what they want to see and how to target ads for **content monetization**. Analyzing **viewer patterns** increases their **revenues**.

Internet of Things (IoT): Big data enhances IoT devices, which generate **continuous data**. Analytics on this data provides **personalized customer experiences**. Big data is essential for **unlocking IoT's full potential** by giving meaningful insights from massive data.

Manufacturing: Big data helps manufacturers make **better products and smarter decisions**. It predicts when machines need maintenance (**predictive maintenance**), improves how products are made (**cheaper and better**), and acts like a **smart assistant** for the whole process.

Government and Big Data: Big data analytics helps governments **understand citizens' needs, combat fraud, minimize system errors, and improve operations**. It reduces costs and improves services. By adopting big data systems, governments achieve **efficiency in cost, output, and innovation**.